

Pascal's triangle



1 The 4th row of Pascal's triangle consists of the numbers 1, 4, 6, 4, 1.

- a Write down the numbers in the 5th row of Pascal's triangle.
- b Find the sum of the entries in the 5th row of Pascal's triangle.
- c Rewrite each element of the 5th row of Pascal's triangle as an ${}_nC_r$ coefficient.

2 Complete the following entries in a particular row of Pascal's triangle:

$$1, 8, \text{ }, 56, 70, \text{ }, 28, \text{ }, 1$$

3 Which row of Pascal's triangle would give us the coefficients of terms in the distribution of $(x + y)^7$?

4 State the number of terms in the following distributions:

a $(4x + y)^6$

b $(m + y)^8$

5 Consider the distribution of $(y + k)^4$.

- a How many terms will there be in the distribution?
- b Write the coefficient of the 2nd term.

6 Using the relevant row of Pascal's triangle, determine the coefficient of each term in the following distributions:

a $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$

b $(x + m)^4 = 1x^4m^0 + 4x^3m^1 + 6x^2m^2 + 4x^1m^3 + 1x^0m^4$

c $(5 + b)^5 = 1 \times 5^5b^0 + 5 \times 5^4b^1 + 10 \times 5^3b^2 + 10 \times 5^2b^3 + 5 \times 5^1b^4 + 1 \times 5^0b^5$

d $(ka - b)^6 = 1(ka)^4(-b)^0 + 6(ka)^3(-b)^1 + 15(ka)^2(-b)^2 + 20(ka)^1(-b)^3 + 15(ka)^0(-b)^4 + 6(ka)^0(-b)^5 + 1(ka)^0(-b)^6$

7 Distribute the following using Pascal's triangle:

a $(p + q)^5$

b $(c - d)^8$

c $(y + 3)^4$

d $(u + 4v)^6$

e $\left(1 + \frac{4}{y}\right)^3$

8 Evaluate the following:



a $\binom{6}{5}$

b $\binom{14}{1}$

c $\binom{10}{9}$

d $\binom{8}{0}$

Binomial theorem

9 Consider the distribution of $(a - 4)^5$. Will the 2nd term be positive or negative?

10 Complete the following distribution:

$$(4p + 3q)^3 = {}_3C_0 (4p)^0 (3q)^3 + {}_3C_1 (4p)^1 (3q)^2 + {}_3C_2 (4p)^2 (3q)^1 + {}_3C_3 (4p)^3 (3q)^0$$

11 Consider the binomial $(4x + 3y)^4$.

a State the first term in the distribution.

b State the last term in the distribution.

12 Which two terms in the distribution of $(u + v)^{11}$ have a coefficient of ${}_{11}C_9$?

13 Find the specified term for the following distributions:

a Fifth term of $(b + 2)^7$.

b Fifth term of $(c + d)^9$.

c Fifth term of $(3u - v)^6$.

d Tenth term of $\left(u + \frac{1}{2}\right)^{12}$.

e Seventh term of $\left(\frac{3y}{2} - \frac{2}{3y}\right)^{10}$.

f Eleventh term of $(2x + y^2)^{13}$.

g Sixteenth term of $(x - y^5)^{19}$.

h Middle term of $(4x^2 + 3y^3)^8$.

i Middle term of $(-3x^{-3} + 2y^{-2})^4$.

j Constant term of $\left(2y - \frac{1}{y^3}\right)^8$.

14 Use the binomial theorem to distribute the following expressions:

a $(1 + x)^4$

b $(2 + b)^5$

c $(p + q)^5$

d $(c - d)^8$

e $(u + 4v)^6$

f $(y + 3)^4$

g $(ka - b)^6$

h $(y - 5)^3$

i $\left(y - \frac{1}{3}\right)^3$

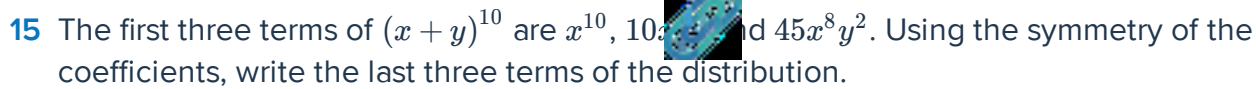
j $\left(2 + \frac{y}{2}\right)^3$

k $(3y + 2r)^3$

l $\left(a - \frac{1}{a}\right)^3$

m $(u^2 + 3v^2)^3$

n $\left(1 + \frac{4}{y}\right)^3$



a Find the value of n . **b** Find the 9th term.

18 Find the coefficient of a^3b^6 in the distribution of $\left(2a - \frac{b}{2}\right)^9$.

20 Find:

c The term in $(u + v)^{11}$ that contains v^8 . **d** The term in $(u + v)^{15}$ that contains u^5 .

22 Find the value of n such that the distribution of $(a + b)^n$ contains the term $84a^6b^3$.

24 By considering $(x + y)^4 = (x + y)(x + y)(x + y)(x + y)$, in how many ways can three factors of x and one factor of y be chosen from the distribution to form the term which contains x^3y^1 ? Write your answer in the form ${}_nC_r$.

- State the coefficient of the 3rd term in the form ${}_nC_r$.
- Evaluate the coefficient of the 3rd term.
- By considering the symmetry of ${}_nC_r$, which other term has the same coefficient as the 3rd term?

26 Complete the following distribution by determining the missing power and binomial coefficient of the $(k + 1)$ th term:

$$(3 - x)^8 = {}_8C_0 \times 3^8 (-x)^0 + {}_8C_1 \times 3^7 (-x)^1 + \dots + \mathbb{I} \times 3^{8-k} (-x)^{\mathbb{I}} + \dots + {}_8C_8 \times 3^0 (-x)^8$$

27 Consider the binomial series:

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

- a** Use the first four terms to approximate $(1.03)^{-2}$ to the nearest thousandth.
- b** Use the first four terms to approximate $(1.05)^{\frac{3}{4}}$ to the nearest thousandth.
- c** Use the first five terms to approximate $\frac{1}{(1.06)^2}$ to the nearest thousandth.